



Modeling health equity in active transportation planning

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ABSTRACT

Quantifying the public health impacts of transportation plans provides stakeholders with information that can aid in evaluating alternative transportation scenarios. Active transportation can improve public health through increased physical activity. At the same time, increasing walking and cycling can affect the risk of traffic injury or increase exposure to air pollution. The Integrated Transport and Health Impacts Model (ITHIM) has been applied around the world to understand changes in public health that will result from increases in active travel. However, the model does not provide disaggregate information required to evaluate the social equity implications of such changes. Health benefits are typically reported as a single value (e.g., total reductions in deaths) at the county or regional level. In this work, we draw from several data sources to report demographically explicit (i.e., race/ethnicity and income) results from an ITHIM implementation developed for the Sacramento region in California. This disaggregation is helpful because travel behavior and health outcomes are affected by race/ethnicity and its correlates (e.g., residential location) and planning agencies are required to ensure that their policies and projects are not discriminatory. The results show people of color are generally expected to experience greater overall health benefits than non-Hispanic whites. In addition, the disaggregated results indicate that there is a substantial variation between counties and demographic groups, providing decision makers with the information needed to target interventions to achieve desired outcomes for disadvantaged populations. They also pave the way for further spatial disaggregation.

1. Introduction

Contemporary transportation systems are unsustainable and inequitable, contributing to global climate change, emissions of hazardous air pollutants, and resulting in often-fatal traffic collisions (Chakraborty, 2006, 2009; Lee et al., 2017; Lucas, 2004, 2012). Policies and planning efforts aimed at achieving sustainability can take a number of forms. Technological solutions for sustainable transportation seek to minimize the per-mile impacts of driving by using less carbon-intensive fuels or vehicles (Sperling and Gordon, 2009).

More ambitious comprehensive solutions focus on creating neighborhoods and cities conducive to walking, cycling, and public transit. These strategies can reduce greenhouse gas emissions and exposure to harmful pollutants, improve human health by

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increasing physical activity, and reduce social exclusion by increasing access to amenities for transportation-disadvantaged populations who have few options due to economic or physical constraints (de Hartog et al., 2010; Lindsay et al., 2011). But creating such places can also lead to increased local air pollution exposure and change the risk of vehicle, pedestrian, and bike collisions (Stone et al., 2007; Schweitzer and Zhou, 2010; Ewing and Dumbaugh, 2009). The distribution of resultant risks and benefits can vary across population groups and space (Schweitzer and Valenzuela, 2004; Cubbin and Smith, 2002; Karner et al., 2018). Research on transportation equity has sought to understand the extent to which different demographic groups receive fair shares of the benefits of transportation investments without being unduly burdened by the negative effects of infrastructure proximity (Karner et al., 2018; Karner, 2016; Martens et al., 2012).

Understanding these comprehensive and distributional impacts is of particular importance in California, where statewide, regional, and local agencies and jurisdictions are working to implement the Sustainable Communities and Climate Protection Act of 2008 (commonly known as SB 375). SB 375 requires the state's metropolitan planning organizations (MPOs) to adopt a Sustainable Communities Strategy (SCS) as a new component of long-range regional transportation plans (RTPs). An SCS must describe and evaluate a combined transportation, land use, and housing plan that meets a per capita greenhouse gas emissions reduction target. MPOs have generally evaluated the performance of multiple SCS scenarios prior to adopting a preferred plan (Barbour and Deakin, 2012).

While clear links have been made between travel behavior and health outcomes, integrating health impacts into transportation decision making remains challenging because of a lack of readily available tools, data, and methods familiar to transportation planners. In this research, we present a real-world health impact assessment for the six-county Sacramento Area Council of Governments (SACOG) region in northern California aimed at helping inform implementation of the region's new long-range transportation policies. We draw from several data sources to report demographically explicit (i.e., race/ethnicity and income) results, providing decision makers with the information needed to ensure equitable outcomes.

2. Literature review

Because the impacts of a particular transportation project or plan on health and well-being are complex and situation-specific, it is difficult to apply general relationships between transportation and health to project and plan evaluation in practice. Health impact assessment is an approach that can be used to estimate the health outcomes (e.g., changes in morbidity and mortality) expected to result from specific policies and practices that do not have health as their primary goal (Ross et al., 2014). Different methods can be used to assess health impacts, including benefit assessment (e.g. Grabow et al., 2012; Olabarria et al., 2013), cost-benefit assessment (e.g., Gotschi, 2011; Lindsay et al., 2011), risk assessment (e.g., Stipdonk and Reurings, 2012; Schepers and Heinen, 2013) and comparative risk assessment (CRA) (e.g., Edwards and Masonm, 2014; Xia et al., 2015).

The Integrated Transport and Health Impact Model (ITHIM) has been widely applied around the world to provide actionable information that decision makers can use when weighing the relative health benefits and costs of alternative transportation and land use futures (e.g., Maizlish et al., 2013; Maizlish et al., 2017; Woodcock et al., 2009; Woodcock et al., 2013). ITHIM was initially developed by Woodcock et al. (2009) at the Centre for Diet and Activity Research at Cambridge University. It was intended to assess transport-related public health impacts arising from three exposure pathways—air quality, traffic collisions, and physical activity—using CRA methods that observe existing rates of morbidity and mortality under a baseline scenario and then shift those observations based on estimated changes in exposure to risk of illness and death.

ITHIM is not standard off-the-shelf software; rather, a local implementation of ITHIM must be calibrated to a particular geographic area using local estimates of baseline health burdens and exposures and embedding key CRA relationships in a suitable computing platform (using, for example, spreadsheet software, R, or Python). After such calibration, ITHIM provides a powerful tool that can be used to inform transportation decision making and project prioritization. Maizlish et al. (2013) applied ITHIM to visionary transportation and land use scenarios in the San Francisco Bay Area, determining that the potential public health benefits resulting from increased physical activity far outweighed those from air quality improvements and changes in collision rates. In later work, the authors applied ITHIM to quantify the health outcomes of preferred regional transportation plan scenarios in the five most populous California MPO regions (Maizlish et al., 2017). Götschi et al. (2015) used ITHIM to estimate the health benefits from changes in physical activity that would arise if adults in urban areas in England and Wales adopted travel patterns of the Netherlands, California, and Switzerland, in order to quantify and compare health outcomes between those settings. The results demonstrated that differences between high-income regions in population travel patterns have important implications for public health. Sá et al. (2017) applied ITHIM to evaluate the health impacts of different travel pattern scenarios for the adult population in the city of São Paulo, Brazil. The results showed that major health benefits could be achieved by shifting travel patterns towards more sustainable transport, including active travel modes. This research also demonstrated that ITHIM could be calibrated and applied in developing cities. These prior ITHIM applications have indicated that the model can be used to quantify the health impacts of alternative transportation scenarios. In general, the ITHIM results demonstrate that increasing rates of active travel contribute to significant health benefits.

Prior implementations of health impact assessments, including ITHIM, have generally presented aggregate regional results, using a single value to represent vast geographic areas. Such results do not shed light on whether expected improvements/declines in health are equitably distributed across sub-regional areas or population groups. Existing studies have, to some extent, recognized the potential for disparities in population health, but they most often focus on the effects of age and sex (e.g. Rojas-Rueda et al., 2011; Woodcock et al., 2014). Only a limited number of studies have considered the effects of transportation plans on health equity for different racial and ethnic populations. For example, Lindsay et al. (2011) compared the ethnicity-specific mortality benefit that

would arise if short trips were undertaken by non-motorized modes in New Zealand. They found greater health benefits for the Maori and Pacific Islander minorities than the overall population. However, the authors assumed that changes in transportation behavior that result from policy changes would be identical across all population groups. This assumption implies that variation in expected health benefits is only related to differences in baseline mortality rates, which is higher among these minority groups. But the behavioral effects of transportation plans are likely to vary across population groups; that study was not able to capture this variation.

Disaggregate health assessments can be used to demonstrate whether different population groups share equitably in the benefits and burdens of transportation investments, which is required in the United States by federal law and executive agency guidance issued in the wake of Title VI of the 1964 Civil Rights Act (Karner and Niemeier, 2013). Planning agencies that receive federal funds must ensure that people of color and low-income people are not denied the timely and equitable receipt of the benefits of public investments (Marcantonio et al., 2017; Karner, 2016).

Ensuring equitable health outcomes from transportation plans and investments is critical for addressing the often-disparate effects of transportation infrastructure on different groups. Communities of color and low-income households often bear greater air pollution burdens due to their proximity to busy roads (Rowangould, 2013; Tian et al., 2013; Schweitzer and Valenzuela, 2004; Walker, 2010; Chakraborty, 2006, 2009) in addition to facing worse sidewalk quality (Franzini et al., 2010; Kelly et al., 2007), while communities with low socioeconomic status also often face greater risks of traffic injury (Cubbin and Smith, 2002; Laflamme and Diderichsen, 2000; Morency et al., 2012).

Despite these existing inequities, active transportation plans in the US often lack meaningful consideration of equity (Lee et al., 2017). This is problematic because although encouraging active travel can reduce the overall negative (and increase the positive) environmental and health effects of transportation systems, projects and plans that emphasize a shift to non-auto travel may fail to mitigate or even perpetuate existing disparities. For example, residents of high-poverty areas containing concentrations of people with low educational attainment are less likely to receive new pedestrian and bicycle projects across the US (Cradock et al., 2009). Low-income residents can also be priced out of the neighborhoods around light rail stops (Luckey et al., 2018). Similarly, in Albuquerque New Mexico, a smart-growth-oriented regional plan that would reduce region-wide air pollution emissions would increase some residents' typical pollution exposures, worsening disparities in the impacts borne by low-income households (Tayarani et al., 2016). Although these types of transportation equity challenges are widely recognized, generalizable quantitative approaches for evaluating the variation of health outcomes across population groups that will result from proposed transportation plans are lacking.

In addition to overlooking equity, transportation health impact assessments have typically evaluated simplistic hypothetical changes in travel behavior, for example assuming that all automobile trips are replaced with bike share trips (Rojas-Rueda et al., 2011) or that short automobile trips are replaced with non-motorized modes (Rabl and de Nazelle, 2012). Such studies do not account for the nuances of policy implementation and do not consider likely heterogeneity in traveler responses to the policy or transportation plans. Further, changes are uniformly applied across the entire population and do not consider variation in travel behavior across demographics. Activity-based travel demand models use activity participation as the basis for estimating travel patterns. They are better suited to represent the effects of investments, policies, and other changes on travel behavior than traditional four-step travel demand models (Castiglione et al., 2014).

The purpose of this work is to address the above-noted gaps in the existing health impact assessment literature by developing an approach to conducting demographically explicit analysis of the health impacts of transportation plans. The effects of these plans on health equity have not been closely examined, partly because of a lack of generalizable quantitative approaches. In this study, we develop and apply ITHIM physical activity and traffic injury modules to assess the health equity of transportation plans in the SACOG region for different race/ethnicity and income groups. This is important because regional transportation plans allocate federal transportation dollars, and as such are subject to federal law and executive agency guidance related to equity. We synthesize data across a wide range of sources and use “data fusion” techniques to impute missing demographic information as needed. The model is tested using simulated travel behavior data estimated by SACOG's activity-based travel demand model, SACSIM15, which underlies their 2016 Metropolitan Transportation Plan/Sustainable Communities Strategy (MTP/SCS). In order to compare results across differently sized geographic units and demographic groups, results are normalized by age and population.

3. Study scope

SACOG has been advancing the frontiers of transportation modeling and performance assessment for some time (SACOG, 2004) and a number of regional stakeholders have expressed interest in modeling the health impacts of transportation policies and plans in order to inform their design. Additionally, a previously developed version of ITHIM includes the Sacramento region (Maizlish et al., 2017), providing a starting place for the enhanced ITHIM functionality developed here.

SACOG used SACSIM15 to estimate travel behavior for the present and for changes that will occur in the future, including three alternative scenarios under their 2016 MTP/SCS. The plan must meet the mandates of SB 375 requiring per-capita reductions in future GHG emissions resulting from changes in travel behavior (SACOG, 2016). Table 1 shows county-level demographics for the six counties in the SACOG region: El Dorado, Placer, Sacramento, Sutter, Yolo, and Yuba.

In this study, we evaluate the changes in health risks due to changes in physical activity and traffic injury that will result from future years and alternative scenarios under the 2016 MTP/SCS. In the physical activity module, health outcomes and changes in travel behavior are modeled separately for age-gender-race/ethnicity and age-gender-income categories in each county. In the traffic injury module, income data were unavailable and injury data were too sparse to disaggregate for all four race/ethnicity categories. Thus, traffic injury health outcomes are modeled separately for two race/ethnicity groups (non-Hispanic white and “people of color,” which includes Black, Hispanic, and other residents) in each county. In general, we use the travel behavior of individuals simulated

Table 1
Sacramento region demographics (total population: 2.27 million).

Population – grouped by race/ethnicity					
	Count	White	Black	Hispanic	Other
El Dorado	151,639	79.6%	0.3%	11.7%	8.4%
Placer	346,984	76.1%	0.8%	12.6%	10.5%
Sacramento	1,402,302	47.9%	9.3%	21.6%	21.2%
Sutter	93,877	53.6%	1.9%	27.4%	17.1%
Yolo	197,101	49.1%	2.4%	31.5%	17.0%
Yuba	73,552	52.8%	1.8%	28.1%	17.3%
Population – grouped by household income (\$ per year)					
	Count	Quartile 1 < \$32,000	Quartile 2 \$32,000–\$62,090	Quartile 3 \$62,090–\$105,000	Quartile 4 > \$105,000
El Dorado	151,639	15.1%	20.4%	27.1%	37.8%
Placer	346,984	16.4%	19.7%	26.5%	37.5%
Sacramento	1,402,302	26.9%	26.3%	25.0%	21.7%
Sutter	93,877	28.0%	31.8%	23.5%	17.4%
Yolo	197,101	29.9%	23.3%	21.6%	25.2%
Yuba	73,552	31.5%	32.3%	24.0%	12.3%

Source: 2012 American Community Survey 1-year estimates. White, Black, and Other categories include non-Hispanic residents of each race while the Hispanic category captures Hispanic residents of any race. Income quartiles are defined using the region's income distribution.

from SACSIM15 as the basis of our physical activity and traffic injury exposure measures. Individuals are variously aggregated into groups according to their county of residence, race/ethnicity and household income category. Health outcome results are ultimately presented at the county level.

4. Methods and data

The fundamental methodological approach employed by ITHIM for both the physical activity and traffic injury modules is CRA. The relative risk (RR) in CRA is defined as “the ratio of the probability of developing the condition (e.g., a disease or injury) if exposed to a certain variable compared with the probability if not exposed” (Mann, 2003, p. 60). CRA can be used to quantify the extent to which disease burden changes when exposure to specific risk factors changes (Vander Hoorn et al., 2004). It is an extension of the population attributable fraction (PAF) formula, as described in Maizlish et al. (2013). Changes in disease burden (DB) for the physical activity and traffic injury exposure pathways are evaluated and then combined as shown in Eqs. (1) and (2):

$$\Delta DB_i = PAF_i \times DB_{i,baseline} \quad (1)$$

$$\Delta DB_{total} = \sum \Delta DB_i \quad (2)$$

where i represents each exposure pathways, i.e., physical activity and traffic injury.

Different calculation methods for PAF are applied for each exposure pathway. For the physical activity exposure pathway, the PAF is estimated as a function of the exposed population and the RR in the baseline and alternative scenarios, which is shown in Eq. (3).

$$PAF_{physicalactivity} = \frac{\int RR(x)P(x)dx - \int RR(x)Q(x)dx}{\int RR(x)P(x)dx} \quad (3)$$

The baseline and alternative population distributions exposed to x are $P(x)$ and $Q(x)$, respectively. The relative risk at a continuous exposure level x , denoted $RR(x)$, was estimated from exposure-response relationships using the physical activity exposure distribution in the baseline population and in the alternative. In ITHIM, physical activity is quantified as the metabolic equivalent of task (MET), which is used to convert time spent in active travel to energy expenditure and thus changes in health outcomes. One MET is defined as the amount of energy expended while at rest and is expressed in the unit $kcal/(kg \cdot h)$ (Ainsworth et al., 2000). Engaging in physical activity results in higher MET values. Calculated MET values are multiplied by the time spent in the activity to arrive at a figure for MET-hours per week, or total energy expenditure.

To determine the appropriate location along the RR– MET-hour gradient, we used a square root function with the relative risk of all-cause mortality for 1 MET-hour equal to 0.94 according to (Woodcock et al., 2009):

$$RR_{exposure} = 0.94^{\sqrt{exposure}} \quad (4)$$

where exposure to physical activity is quantified by MET-hours.

For traffic injury, the PAF was estimated as a function of the baseline total number of injuries and the predicted total number of injuries for a scenario as follows:

$$PAF_{injury} = \frac{\sum Injury_{baseline} - \sum Injury_{future}}{\sum Injury_{baseline}} \quad (5)$$

We formulated the projection of future injuries by multiplying changes in vehicle miles traveled (VMT) for each combination of victim and striking vehicle by the baseline injuries for each race/ethnicity group across each county. This projection method is also applied in other ITHIM applications (Woodcock et al., 2013; Maizlish et al., 2017). A square root function was applied to account for the increase in safety expected to arise from increasing numbers of non-motorized travelers, known as the “safety-in-numbers” effect (Elvik and Bjørnskau, 2017). The victim and striking vehicle modes include pedestrian, bicyclist, motorcycle, car, bus, and truck. The estimation is stratified by severity (fatal and serious injury) and road type (local, arterial and highway). The severity category is used to estimate the *PAF* for different types of health burdens. The *PAF* estimated from fatal injuries is applied to calculate changes in mortality and the years of life lost due to premature death. The *PAF* estimated from serious injuries is applied to years living with a disability. A detailed definition of health burden is included in the following section. Road type serves as a proxy for road geometry, speed and traffic volume. The formulation described below is applied across each severity category, road type, and victim/striking mode combination:

$$Injury_{future} = Injury_{baseline} \cdot Multiplier \quad (6)$$

$$Multiplier = \left(\frac{VMT_{victim, future}}{VMT_{victim, baseline}} \right)^m \cdot \left(\frac{VMT_{striking, future}}{VMT_{striking, baseline}} \right)^n \quad (7)$$

where m and n represent the “safety-in-numbers” parameters for victim mode and striking mode, respectively. Here, we apply 0.5 for all modes, following Maizlish et al. (2017).

Subsequently, we sum the results to generate total predicted injuries for the future year, which is then used in Eq. (5). Finally, the *PAF* is combined with the baseline disease burden to arrive at the change in disease burden.

Data for the Sacramento ITHIM implementation were compiled from a number of sources describing physical activity, travel, traffic injuries and health. An overall schematic of the model structure and required inputs is shown in Fig. 1. The data, their sources, and manipulation are described further below. We implemented this ITHIM application in R (R Core Team, 2013) and drew on earlier R-ITHIM implementations (e.g., Vargo et al., 2017).

4.1. Baseline health data

Health data required for ITHIM include all-cause mortality rates for the Sacramento region and the overall disease burden for the US.

All-cause mortality is used to estimate the local disease burden for physical activity and was taken from the California Department of Public Health (CDPH) vital statistics dataset from 2008 to 2010. These years were chosen because the base year in SACOG’s latest activity-based model, SACSIM15, is 2012 and the CDPH data after 2010 do not include the same set of demographic variables that are included through 2010. An average annual all-cause mortality rate was constructed for each age-sex-race/ethnicity and age-sex-income category, separately for each county. Upon further investigation, mortality rates for the Black population were close to or equal to zero in a number of cases across the region due to the small Black population in many counties. For this reason, we calculated a single region-wide estimate of the annual all-cause mortality rate for the Black population for use in later calculations.

The US disease burden in each age-sex category was extracted from the publicly available Global Burden of Disease (GBD) database (Institute for Health Metrics and Evaluation, 2017). For physical activity, the health burden was based on “all-cause” and for traffic injury the health burden was based on “transport injuries.” Consistent with earlier ITHIM implementations, we used both mortality and disability-adjusted life years (DALYs) as measures of disease burden. DALY estimates are only available at the US level and consider both years of life lost due to premature mortality and the reduction in quality of life caused by years spent living with illness-related disability. They are calculated by summing years living with a disability and years of life lost due to premature death. The US disease burden is taken from GBD 2010 for consistency with the time period of the CDPH vital statistics data.

For physical activity, DALY estimates for the Sacramento region are calculated by scaling US DALYs by the mortality rate in each county over the US mortality rate a whole (estimated as deaths per capita in each age-sex-race/ethnicity and age-sex-income category).

For traffic injury, the baseline health burden estimates are calculated by scaling US health burden of traffic injury from GBD 2010 by the local population over the US population.

4.2. Baseline physical activity data

SACOG used SACSIM15 to provide transportation forecasts for the latest 2016 MTP/SCS and continues to use it for various policy analyses (Bradley et al., 2010). SACSIM15 was calibrated and tested for a base year of 2012.

Transport-related physical activity was quantified using SACSIM and the 2005 California Health Interview Survey (CHIS) was used to quantify non-transport physical activity (UCLA Center for Health Policy Research 2009). Although more recent CHIS data are available, 2005 was the most recent survey year that included occupation type (used to estimate occupational activity) and exercise habits. MET-hours per week are also calculated for occupational and exercise physical activity (non-travel METs) using standard values from Ainsworth et al. (2000) and Maizlish et al. (2013).

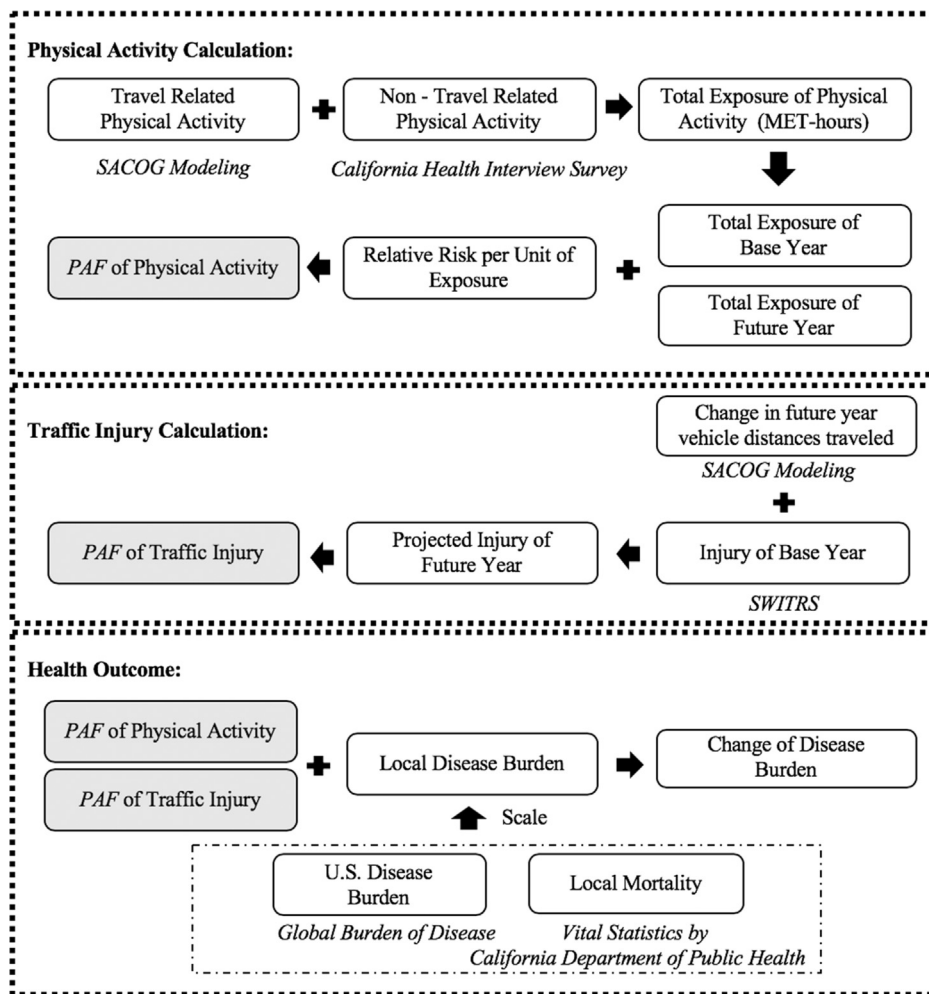


Fig. 1. CRA schematic. Data sources shown in italics.

4.3. Baseline traffic injury data

We extracted 2006 to 2016 traffic injury data for the Sacramento region from the Statewide Integrated Traffic System (SWITRS) (SWITRS, 2018) geocoded by University of California, Berkeley Safety Transportation Resource and Education Center in its Transport Injury Mapping System (TIMS, 2018). These data include the collision location, striking/victim mode, victim race/ethnicity, and the road type associated with each injury. Road type is defined as local, arterial, and highway. Travel modes include bicycle, walk, motorcycle, passenger car, bus and truck. Traffic injuries are categorized as fatal or serious.

These data are used to estimate the baseline rate of injury for each road type/victim mode/striking mode/demographic/county combination. SWITRS includes the race/ethnicity of the victim but not their household income, therefore we estimate mortality and disease burden only by race/ethnicity (not income level) in the traffic injury portion of this analysis. Additionally, because we disaggregate injuries across many dimensions, baseline injuries for some combinations of victim and striking modes for some county/demographic categories equal zero even when there is a nonzero risk (e.g., areas with a risk of 0.1 injuries over a 10-year period would lead to values of zero in most cases). For this reason, we use the region-wide injury rates for road type/victim mode/striking mode/victim demographic combinations where at least one county had a zero.

4.4. Scenario travel data

Travel data include travel distances and times estimated from SACSIM. SACSIM15 was calibrated and tested for a base year of 2012 and was used to forecast travel behavior in 2020, 2027 and 2036 under the adopted 2016 plan. Three alternatives (S1, S2, and S3) to the future year 2036 were also constructed and varied principally by how much housing and transportation choice they created by 2036. The scenarios started with a refresh of the previous MTP/SCS as S2, which was then bracketed with a lower density, high auto investment S1 and a higher density, high multimodal investment S3. In some respects, S2 is similar to the adopted 2016 plan,

Table 2

Description of 2016 MTP/SCS scenarios 1, 2, 3 and adopted 2016 plan (SACOG, 2016).

Transportation inputs	S1	S2	S3	Adopted 2016 plan
New or expanded roads (lane miles, percent increase from 2008)	34%	31%	25%	32%
Transit service (vehicle service hours, percent increase from 2008)	54%	88%	127%	98%
Funding for maintaining and operating the road and highway (\$ in billions)	\$10.9	\$11.5	\$11.0	\$11.5
Funding for maintaining and operating the transit system (\$ in billions)	\$7.5	\$7.9	\$9.6	\$7.9
Funding for bike and pedestrian routes, trails and paths (\$ in billions)	\$2.8	\$2.8	\$3.0	\$2.8
Funding for new or expanded bus and light rail lines (\$ in billions)	\$3.2	\$3.4	\$4.1	\$3.4
Land use inputs	S1	S2	S3	Adopted 2016 plan
Share of growth in “Center and Corridor Communities”				
Percent of new homes	20%	31%	36%	30%
Percent of new jobs	27%	36%	35%	29%
Share of growth in “Established Communities”				
Percent of new homes	29%	25%	27%	26%
Percent of new jobs	57%	49%	53%	52%
Share of growth in “Developing Communities”				
Percent of new homes	47%	42%	36%	42%
Percent of new jobs	15%	14%	12%	18%

although the adopted plan provides more transit service, slightly more new or expanded roads, and channels slightly more job growth into established and developing communities. A detailed comparison is shown in Table 2.

We used SACOG’s SACSIM15 MTP/SCS outputs to estimate the average active transport (walking and cycling) time (minutes per day) and average distance (miles per day) for each demographic group for each of the four analysis years and three scenarios. VMT for each travel mode was estimated for bus, passenger car, truck and motorcycle modes, which is used to predict future-year traffic injuries.

4.5. Data fusion using “hot deck” imputation

In order to conduct the equity analysis, we require travel and health data for each race/ethnicity group and household income group in each county. However, in some data sets, race/ethnicity or income data are missing. For example, SACSIM15 does not include a race/ethnicity variable within its synthetic population. Therefore, we applied hot deck imputation, a type of data fusion method, to impute missing variables as needed. Fundamentally, hot deck imputation seeks to match records drawn from two or more data sources based on other shared variables. Its goal is to incorporate variables from a “donor” dataset into a “recipient” dataset (D’Orazio et al., 2006).

In our case, we used the 2012 American Community Survey Public Use Microdata Sample (ACS PUMS) data 1-year estimates as the donor and SACSIM15 as the recipient. These two data sets share four matching variables: age, gender, income and county of residence. The missing variable in SACSIM15 is race/ethnicity. Similarly, we applied the hot deck method to impute the household income quartile into the CDPH mortality data (recipient) also using the ACS PUMS records with five matching variables: age, gender, race/ethnicity, education level and county of residence. We used the R package “StatMatch” to implement the hot deck method (D’Orazio et al., 2006).

For the traffic injury module, we did not evaluate the victim population by household income, because the victim database does not have income information and other demographic variables are insufficient to perform hot deck imputation.

5. Results

5.1. Regionwide health impacts from physical activity and traffic injury

When evaluating the range of health impacts of a plan or converting health outcomes to economic outcomes, it can be informative to evaluate both DALYs and premature mortality. However, for the purposes of demonstrating the demographically explicit analysis and evaluating the equity of the MTP/SCS in future years and scenarios, we confine this discussion to the expected reductions in premature mortality, as we found that changes in DALYs follow similar trends. Because of this similarity, these latter results are not shown here.

Since each demographic group has a different age/gender profile, we examine age-standardized health outcomes (per 100,000 population). Age standardization applies the health outcomes of a particular population to the population distribution of a larger

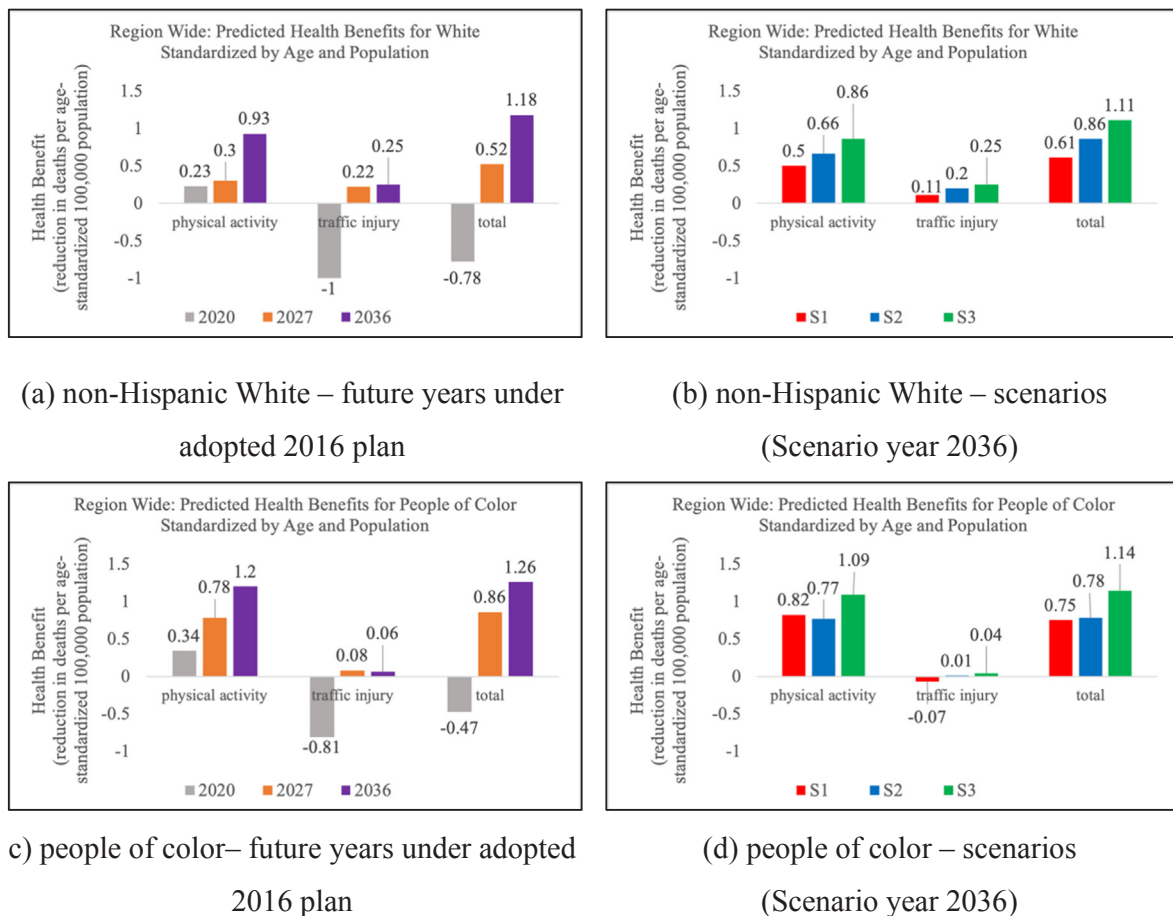


Fig. 2. Predicted health benefits from physical activity and traffic injury compared with baseline year 2012.

reference group so that rates from different locations can be compared independently of their underlying age distribution. We also present results per 100,000 people to allow for comparisons across areas with different total populations. We apply this calculation method for all demographic groups to facilitate a comparison of the change in risks faced by individuals in these populations (rather than a measure of the total expected impact on each population).

Fig. 2 shows the region-wide predicted health benefits from both physical activity and traffic injury compared with base year 2012. Fig. 2(a) and Fig. 2(c) show the results of future year estimates under the adopted 2016 plan for non-Hispanic whites and people of color, respectively. For 2020, both groups exhibit a negative value in death risk reduction (indicating that higher death risks are expected in 2020), which is caused by an increase in traffic injuries. The main cause is an increase in non-motorized travel distances from 2012 to 2020 (11% increase for non-Hispanic white and 12% increase for people of color), which leads to an increase of traffic collision risks that is not offset by the safety in numbers effect.

For 2027 and 2036, both physical activity and traffic injury show reductions in premature death risks. This result arises because active travel time increases and motorized travel distance decreases (see supplemental material). The increase in active travel time in 2027 and 2036 causes the number of traffic injuries for pedestrians and cyclists to increase due to increases in the miles traveled by cyclists and pedestrians (although this is partially mitigated by the safety in numbers effect). At the same time, decreases in motorized travel distance result in a reduction in traffic injuries (due to reductions in injuries to vehicle travelers and reductions in the per-mile risk of injury for active travelers as they encounter fewer vehicles). The net effect is an overall reduction in traffic injury when aggregating across all travel modes.

For three alternative scenarios in Fig. 2(b) and Fig. 2(d), the two groups show different trends. The non-Hispanic white population shows increasing health benefits for all categories (physical activity, traffic injury, and total) from S1 to S2 and S3, which follows the overall trend of greater active travel time and distance. For people of color, the reduction in premature death risk from physical activity in S1 is larger than S2, however, the overall active travel time in S1 is smaller than S2. This is caused by the aggregation of age/gender categories shown here—we estimate the physical activity and health burden in each age and sex category separately, and then aggregate them to output the final results for an age-standardized population. Although the overall active transport time is increasing, in any one age/gender subcategory it may decrease. If a subcategory that exhibits an increase in active travel has a high baseline health burden (e.g., elderly) it will affect the estimated health outcome more than age/gender subgroups with a lower

baseline health burden (e.g., children and young adults), generating results that appear counterintuitive when examining active travel and health in the aggregate.

Comparing race/ethnicity groups, people of color are generally expected to experience a greater health risk reduction benefit from the scenarios analyzed here than the non-Hispanic white population, which is caused by greater changes in physical activity for the former group.

5.2. County-level health impacts from physical activity

We now turn to the spatially disaggregated health outcomes for each county and income group resulting from changes in physical activity. Traffic injury results are not included here because infrequent traffic collisions at the county level may yield less reliable disaggregated estimates of future collisions. On the other hand, physical activity was modeled at the individual level, and total death numbers by county and income are likely to be sufficiently large to analyze changes in active travel health impacts for the majority of the county/income categories (with the possible exception of the two higher income categories in Yuba County and the highest income category in Sutter County. See total death numbers in Supplemental Materials). Baseline death numbers by county and race/ethnicity were not large enough to yield reliable results, so physical activity outcomes by county and race/ethnicity are not presented here.

Fig. 3 shows the standardized health benefits by household income for future years and scenarios compared to base year 2012 for each county in the SACOG region. We extracted the related active travel time for each demographic group. For the purpose of demonstrating the demographically explicit analysis, we confine this discussion to household income (see supplemental material for full results).

As demonstrated in Fig. 3, we observe substantial differences between demographic groups and counties. For example, the lower-income quantiles in El Dorado and Yolo Counties show greater reductions in death risks (deaths per age-standardized 100,000 people) than higher-income quantiles under the adopted plan and for several modeled scenarios. At the same time, the lowest income quantile in Yuba County shows negative death risks avoided in future years under the adopted plan, indicating an increase in future mortality risks.

In Sacramento and Yolo counties, most of the groups (except the highest quantile in Yolo) experience positive health benefits in all years under the adopted plan. Sacramento County is the densest county in the region, and home to the City of Sacramento, the regional job center, which has a walkable downtown and several other traditional pedestrian-friendly neighborhoods. Yolo County is home to the City of Davis, a college town well known for its high rates of cycling. The MTP/SCS indicates that SACOG provides \$2.8 billion for bicycle and pedestrian improvements under the adopted plan, which results in increasing active travel time through infrastructure provision. Sacramento has the greatest increase in cycling time and Yolo has the greatest increase of walking time under the adopted plan (see supplemental material). The convenient pedestrian and bicycle environments are the main reason for the positive health impacts in those two counties. On the other hand, Yuba County shows increases in mortality risks for the lowest-income quantile under the adopted plan, primarily caused by a decrease in active travel time, particularly for cycling and among older travelers (see supplemental material).

Comparing across income categories, in El Dorado and Yolo Counties the lowest income groups receive greater health benefits from plan implementation due to greater increases in active travel time relative to other groups in the same county (see supplemental material). This is likely, in part, because socioeconomically disadvantaged individuals are more likely to use active transport modes, consistent with reports of higher rates of active travel among disadvantaged populations in other studies (e.g. Aytur et al., 2008; Sallis et al., 2009). These groups also experience greater health benefits because of the higher baseline health risks of the travelers that increase their active travel.

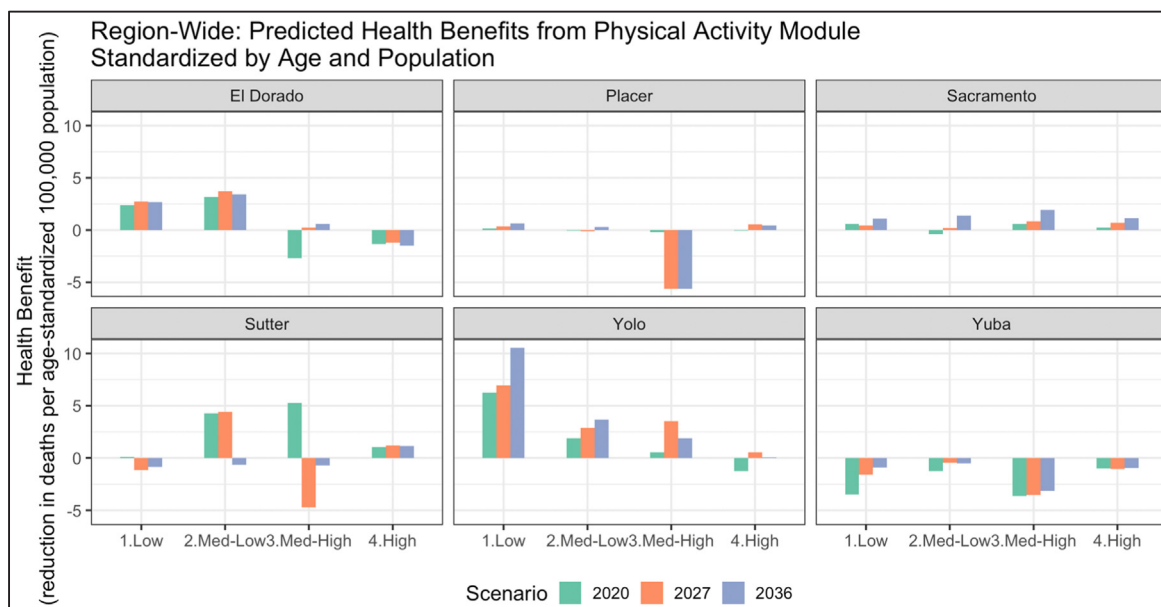
6. Discussion

This study draws from several data sources to estimate the future health effects of changes in physical activity and traffic injury for different race/ethnicity and household income populations in the six counties in the Sacramento region. As expected, for physical activity, we find that the direction of the change in health burden is inversely related to the change in physical activity: increases in physical activity correspond to improved health outcomes and vice versa. Similarly, for traffic injury, the change in health burden is related to the change of distance traveled by each mode that serves as a proxy for traffic collision risk. The magnitude of health outcomes (risk of premature mortality and DALYs) is caused by the magnitude of changes in physical activity and traffic injury, and the magnitude of baseline health burdens.

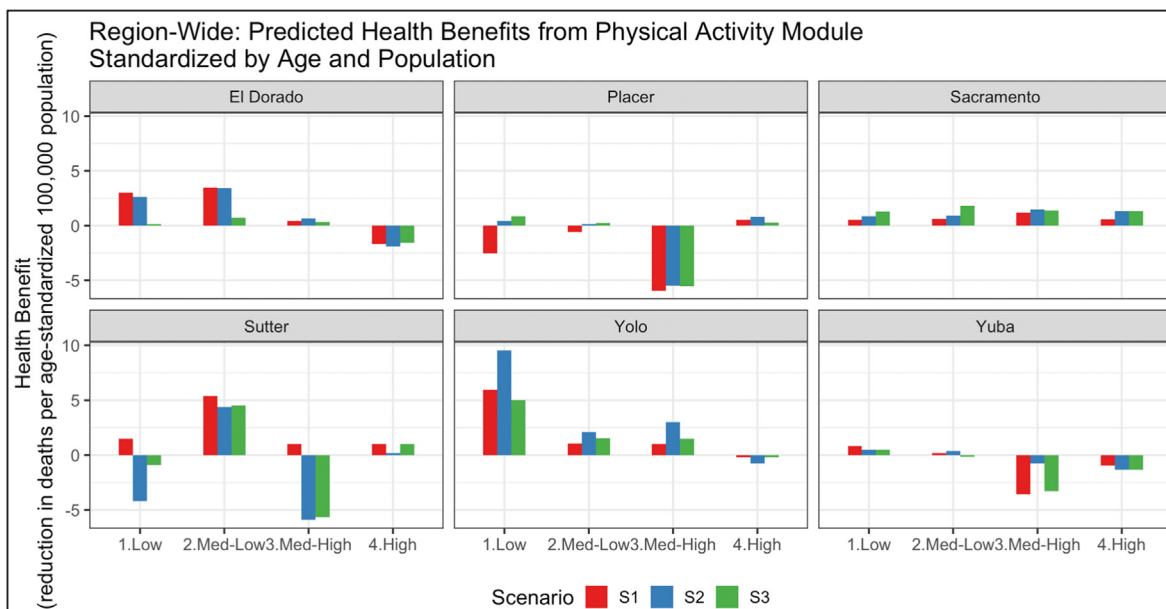
6.1. Social equity analysis

For the region-wide overall health impacts by race/ethnicity, we observe a difference in the expected changes in health burdens for non-Hispanic white residents and people of color. The age-standardized results show that people of color are generally expected to experience greater overall health benefits than non-Hispanic whites, caused by greater changes in physical activity. This result demonstrates the effect of active transport promotion in the 2016 MTP/SCS (e.g., improvements in pedestrian and bicycle facilities) could disproportionately benefit people of color.

When examining the disaggregated physical activity results by race/ethnicity and income, we observe that expected changes in health burdens vary widely across demographic groups and counties. For example, residents of Yolo County show the greatest



(a) future years under adopted 2016 plan



(b) scenarios (scenario year 2036)

* Income quantiles are defined as follows: Low (quantile 1) is <\$32,000/year, Med-Low (quantile 2) is \$32,000 - \$62,090/year, Med-High (quantile 3) is \$62,090 - 105,000/year, and High (quantile 4) is >\$105,000/year.

Fig. 3. Predicted health benefits from physical activity by county, and household income compared with baseline year 2012.

reductions in death risks (as age-standardized deaths per 100,000 people) resulting from changes in physical activity. This result is likely caused by higher rates of walking and cycling than other counties and projected improvements and extension of pedestrian and bicycle routes. The lowest-income residents of Yolo County show a greater relative death risk reduction than other income residents, which means those improvements are more likely to benefit lower socioeconomic status individuals.

Overall, the substantial variation between counties and demographic groups demonstrates the utility of demographically explicit equity analysis, which can be used to ensure that different populations and communities equitably share the costs and benefits of transportation investments. This type of health equity evaluation can inform decision-making and aid in project prioritization. For example, Fig. 3 indicates that low-income residents in Sutter County are not expected to experience the gains realized by other groups. A transportation decision maker faced with this type of finding could then look more closely at what causes this result in this area and consider additional interventions targeting active travel among this population.

It is critical to understand what drives observed differences rather than simply comparing outcomes across groups. For example, it would be important to understand whether increases in physical activity and substantial reductions in death risks for low-income groups are caused by better bike, pedestrian, and transit options, or greater distances to transit stops and reduced automobile ownership. The former implies an overall improvement in accessibility and transportation options, while the latter implies a decrease in accessibility which may compromise health and wellbeing in other ways, e.g., via reduced access to health care and economic opportunities. A closer examination of the transportation plan itself and the modeled travel behavior could provide insight into the underlying causes of observed differences.

6.2. Strengths and limitations

This study used hot deck imputation to infer missing demographic variables to support a demographically resolved analysis. This approach results in robust estimates of outcomes for the populations examined. At the same time, this method glosses over complexities in behavior or baseline health burdens that the underlying data and matching variables fail to capture. That is, to the extent that travel behavior varies by race/ethnicity independently of the match variables used to impute the missing demographics in the SACSIM data (age, gender, income, county) this analysis would fail to capture that nuance. Similarly, the physical activity and traffic injury assumptions and relationships are somewhat simplified in order to conduct a broad analysis at the regional scale. This is consistent with previous ITHIM work (e.g., Maizlish et al., 2017; Maizlish et al., 2013; Woodcock et al., 2013), and does not reflect outcomes at small spatial scales (e.g. the neighborhood-level), or the granularity that has been explored in some research (e.g., the effects of localized safety interventions such as converting traditional intersections to roundabouts (Persaud et al., 2001)). Furthermore, the functions in ITHIM are applied uniformly across demographic groups; this analysis does not capture any variation in the relationship between physical activity and health that may be tied to specific race/ethnicity or income-based populations. Similarly, variation in input data that may occur at the sub-county level is not addressed here. Additionally, in some cases the available data for a subpopulation are sparse and therefore noisy, so some spatial aggregation of data was necessary, e.g., for Black residents in the study area. Finally, the analysis assumes that several characteristics of the future population and risks are the same as in recent years (e.g., baseline health and activity data, baseline injury data, and relationships between demographic variables used to impute missing data) due to the difficulty in predicting these characteristics.

It is important to note that this analysis accounts for the benefits of increased physical activity without accounting for the risks due to exposure to air pollution. Previous studies indicate that in general the benefits of increased physical activity outweigh the risks posed by the additional exposure to air pollution that active travel presents except in areas where air pollution concentrations greatly exceed typical levels found in the US (Tainio et al., 2016; de Nazelle et al., 2015). In a comprehensive review of studies that evaluate the population-level health impacts of shifts to active travel, Mueller et al. (2015) demonstrated that the total benefits of physical activity greatly outweighed the risks posed by traffic injury and air pollution exposure of active travelers in all studies reviewed, while most studies similarly found that air pollution benefits to the general population were small relative to physical activity benefits (with the exception of two studies that assumed substantial reductions in vehicle travel (Dhondt et al., 2013; Grabow et al., 2012)). Consistent with these earlier studies, we also find that the benefits of physical activity outweigh the risks of traffic injury. While the health effects of exposure to ambient pollution may be important in the Sacramento region, the change in air quality risks posed to active travelers and the general population are likely to be modest relative to other risks due to the relatively small mode shifts assumed in this analysis.

7. Conclusions

This work has demonstrated the ability of a modified version of ITHIM to assess the distribution of changes in health outcomes across different population groups. The modified model provides disaggregated results by race/ethnicity and income, consistent with the requirements of US federal law and guidance related to civil rights and environmental justice. The distribution of outcomes of a transportation plan, policy, or project can be incorporated into broader evaluations of multiple effects or monetized for incorporation into cost-benefit analyses, leading to more equitable and sustainable transportation planning decisions. It also points to the directions for future scholars to pursue to further refine and apply the model.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.trd.2019.01.011>.

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